

A Causal Probabilistic Transformer

An impossibility theorem, a directed reformulation,
a graph-faithful output mechanism, and the defect analysis

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Abstract

These notes develop, in three parts, a causal (decoder / autoregressive) extension of the Probabilistic Transformer (PT) of Wu & Tu. **Part I** proves an impossibility theorem: for any single differentiable energy over mean-field marginals, *causality*, *single global energy*, and *contextuality* cannot hold simultaneously; any two are compatible. The proof is self-contained, states every regularity assumption at its point of use, and includes an explicit counterexample showing that a naive (trajectory-level) formalisation of causality makes the theorem *false* — so the correct formalisation is architectural. This settles that no redefinition of the potential function can work inside one energy: the model class must change. **Part II** gives the minimal principled change of model class: a directed chain of conditional CRFs, inferred by variational filtering. The acausal message is never born, each step is an honest gradient field, the reduction to masked scaled dot-product attention is exact, the symmetry assumption on **T** becomes unnecessary, and the KV cache falls out of the model. A second construction (per-prefix re-inference) is given for contrast. **Part III** designs the output mechanism as part of the probabilistic model: the words are promoted to variables, prediction is inference over a latent next-slot label with no unary factor, and the mapping to the vocabulary is the word–label factor itself. Weight tying, the learned mask embedding, two-stream attention, and a mixture-of-softmaxes readout are then *derived* rather than posited. All six verification checks are carried out explicitly, and the remaining approximation-level defects are stated rather than glossed. **Part IV** attacks those defects rather than restating them. It quantifies the label bottleneck and relocates the true risk upstream (the mean-field readout’s bottleneck equals the baseline’s own softmax bottleneck at matched width); gives principled in-graph widenings (factored labels with a product-of-mixtures readout; the paper’s global variables as a graph-faithful feed-forward); proves a contraction bound on the prior/posterior divergence with an embedding-surprisal kick; shows that sharing the attention pattern between prediction and encoding is a legal warm-start schedule; inverts an earlier recommendation (the exact tree readout becomes the mainline, removing the query stream); corrects the cost accounting to roughly half a transformer layer; proves the dropped explaining-away term identical to Part II’s and derives a fixed-lag smoothing repair; and specifies the localisation-ladder experiment to run first.

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Notation and conventions

We follow Wu & Tu. A sentence has length n ; Z_i is the discrete latent label of word i over a label set of size d ; $H_i^{(c)}$ is the dependency head variable of word i in channel $c \in \{1, \dots, h\}$. The unary potential is $\phi_u(Z_i) = \exp(\mathbf{S}_{w_i, Z_i})$ with $\mathbf{S} \in \mathbb{R}^{|\mathcal{V}| \times d}$, and the ternary potential is

$$\phi_t(H_i^{(c)}, Z_i, Z_j) = \begin{cases} \exp(\mathbf{T}_{Z_i, Z_j}^{(c)}) & H_i^{(c)} = j, \\ 1 & \text{otherwise.} \end{cases}$$

MFVI (in its extended entropic Frank–Wolfe form, Appendix A of the paper) updates each marginal by

$$Q_x \propto \exp\left(-\frac{1}{\lambda_x} \frac{\partial E}{\partial Q_x}\right), \quad \lambda_x > 0,$$

with $\lambda_Z = 1$, $\lambda_H = 1/d$ by default. We write $Q_i(\cdot)$ for the marginal of Z_i and $Q_{ic}(\cdot)$ for that of $H_i^{(c)}$. Throughout, “term 1” and “term 2” refer to the two summands of the message $G_i^{(t)}(a)$ in Eq. (4) of the paper: term 1 is weighted by $Q_{ic}(j)$ (“who i points to”), term 2 by $Q_{jc}(i)$ (“who points to i ”).

Part I — The impossibility theorem

1 Statement, and two formalisation decisions

The informal claim is: *a causal, contextual inference scheme cannot be the gradient field of a single energy*. Before proving it, two decisions must be made explicit, because both are places where a probing reader will (rightly) apply pressure, and one of them changes the statement.

Decision 1: where the words enter

One might state the theorem with “a word w_j enters only through its initialisation $Q_j^{(0)} \propto \exp(\mathbf{S}_{w_j, \cdot})$ ”. In actual PT this is not quite the case: w_j also enters through the unary term of E , so that $\mathbf{S}_{w_j, \cdot}$ reappears in *every* update of Z_j , not only at $t = 0$. We therefore assume the more general *word-locality* condition (A3 below): w_j may enter through $Q_j^{(0)}$ *and* through an additive term of E that depends on the block Q_j alone. The initialisation-only version is the special case $u_j \equiv 0$.

This assumption is load-bearing rather than cosmetic. If E were allowed to contain word-nonlocal terms, the theorem would be **false**: the energy

$$E(Q; w) = - \sum_i \sum_a Q_i(a) \left(\mathbf{S}_{w_i, a} + \sum_{k < i} \mathbf{S}'_{w_k, a}{}^{(i, k)} \right)$$

is a single smooth energy, is causal, and is contextual — because it conditions on the prefix *directly*, bypassing inference altogether. The theorem is about context that is *mediated by inference over the marginals*; (A3) is the formal expression of exactly that scope.

Decision 2: what “causality” means

Consider the natural, literal notion:

(C1) trajectory causality. For all $i < j$, all schedules and all t : $Q_i^{(t)}$, viewed as a function of the words, does not depend on w_j .

(C1) is a *finite* family of conditions: the vocabulary is finite, so the trajectories visit a constrained set of states. The graph-theoretic conclusion we want (“ Z_i and Z_j are disconnected”) is an *identical-vanishing* condition. These do not coincide, and the gap is real: Proposition 8.1 constructs an explicit polynomial energy (two positions, binary labels) satisfying *single energy*, (C1) and *contextuality* simultaneously. The acausal coupling exists in E , but its coefficient is tuned to vanish at every state the dynamics can reach.

Hence the theorem *as literally stated with (C1) is false*, and no proof of it exists. The correct and provable formalisation is architectural:

(C2) network causality. For every schedule, every t , and every node x (*including* head variables): $Q_x^{(t)}$, as a function of an *arbitrary* interior initialisation, does not depend on the initialisation blocks of nodes at strictly later positions.

(C2) is what masked attention satisfies — for all parameter values, all activations, all inputs — and it is what anyone means by “a causal architecture”. It is also what Approach A (hand-deleting term 2) satisfies. Everything below proves **SE** $\wedge$ **C2** $\Rightarrow$ $\neg$ **Contextuality**, unconditionally; that **C2** $\Rightarrow$ **C1**; that all three pairs of properties are realisable; and that (C1) cannot replace (C2).

2 Setup and standing assumptions

Nodes and state space. Positions $1, \dots, n$. Node set $\mathcal{V} = \{z_1, \dots, z_n\} \cup \{h_{ic} : i \in [n], c \in [h]\}$, with $\pi(x)$ the position of node x (so $\pi(z_i) = \pi(h_{ic}) = i$). Node x carries a distribution over a finite domain of size d_x ($d_x = d$ for label nodes; $d_x = |\mathcal{D}_i|$ for head nodes, with $\mathcal{D}_i$ arbitrary). Write $V = \prod_x \mathbb{R}^{d_x}$ for the ambient space, Δ_x for the simplex of node x and Δ_x° for its relative interior, and

$$\mathcal{X} = \prod_x \Delta_x, \quad \mathcal{X}^\circ = \prod_x \Delta_x^\circ.$$

Tangent directions: $T_x = \{u \in \mathbb{R}^{d_x} : \mathbf{1}^\top u = 0\}$, with orthogonal projector $\Pi_x = I - \frac{1}{d_x} \mathbf{1}\mathbf{1}^\top$ onto T_x (so $\Pi_x u = u$ for $u \in T_x$). A function f on $\mathcal{X}^\circ$ *does not depend on the y -block* if $f(\bar{P}) = f(P')$ whenever $P, P' \in \mathcal{X}^\circ$ differ only in the Q_y coordinates.

Assumption 1 (Regularity). $E \in C^2(U)$ for some open convex $U \supseteq \mathcal{X}$ in V .

Used for: symmetry of the Hessian (Schwarz/Clairaut) and segment integration. The C^2 hypothesis is genuinely needed and not a technicality: mere existence of the mixed partials does not imply their equality, as the classical example $f(x, y) = xy(x^2 - y^2)/(x^2 + y^2)$, $f(0, 0) = 0$, shows ($\partial_x \partial_y f(0, 0) \neq \partial_y \partial_x f(0, 0)$). The entire theorem runs through that symmetry.

Assumption 2 (Updates). A *schedule* is a sequence $\sigma = (B_1, B_2, \dots)$ of non-empty subsets of $\mathcal{V}$. At step t , every $x \in B_t$ is replaced by

$$Q_x^{(t)} = \text{softmax} \left(-\frac{1}{\lambda_{x,t}} \nabla_x E(Q^{(t-1)}) \right), \quad \lambda_{x,t} > 0,$$

and all other nodes carry their values over.

This covers synchronous, asynchronous and mixed schedules, any number of iterations, and per-node, per-step message weights.

Assumption 3 (Word-locality). $E(Q; w) = E_0(Q) + \sum_{i=1}^n u_i(Q_i; w_i)$, where E_0 does not depend on w and each u_i depends only on the block Q_{z_i} . (In PT, $u_i = -\langle \mathbf{S}_{w_i, \cdot}, Q_i \rangle$.) No other part of the scheme depends on w .

Assumption 4 (Initialisation). $Q_{z_i}^{(0)} = \iota_i(w_i) \in \Delta^\circ$ (e.g. $\text{softmax}(\mathbf{S}_{w_i, \cdot})$), and head initialisations are word-independent interior points (e.g. uniform).

Lemma 2.1 (Interiority). *Every iterate lies in $\mathcal{X}^\circ$.*

Proof. Initialisations are interior by Assumption 4, and softmax outputs are strictly positive. Hence all gradient evaluations occur at points of $\mathcal{X}^\circ$. $\square$

Definition 2.2 (The three properties). Fix a scheme (update maps together with an initialisation family).

- **SE** (single global energy): the updates have the form of Assumption 2 for one common E satisfying Assumptions 1 and 3.
- **X** (contextuality): there exist i, t , a schedule, and words w, w' differing only in a coordinate $k \neq i$, with $Q_{z_i}^{(t)}(w) \neq Q_{z_i}^{(t)}(w')$.
- **C1, C2**: as in Section 1.

3 Gauge reduction: what an update actually reads

Lemma 3.1 (Softmax reduction). *Fix a node x and $\lambda > 0$. For $P, P' \in \mathcal{X}^\circ$,*

$$\text{softmax} \left(-\frac{1}{\lambda} \nabla_x E(P) \right) = \text{softmax} \left(-\frac{1}{\lambda} \nabla_x E(P') \right) \iff \Pi_x \nabla_x E(P) = \Pi_x \nabla_x E(P').$$

Consequently, for $y \neq x$: the x -update depends on the y -block somewhere on $\mathcal{X}^\circ$ if and only if the centred field $\Pi_x \nabla_x E$ does.

Proof. $\text{softmax}(g) = \text{softmax}(g + c\mathbf{1})$ for every $c \in \mathbb{R}$, and softmax restricted to T_x is injective: if $\text{softmax}(g) = \text{softmax}(g')$ then $g - g' = c\mathbf{1}$, and centred vectors force $c = 0$. Multiplication by $-1/\lambda$ is a bijection of T_x . $\square$

This is the precise sense in which an update reads *only* the tangential content of the gradient. It is also the formal home of the shift-invariance loophole, which we now close by *definition* rather than by a later patch.

Definition 3.2 (Reading relation and interaction graph). For $x \neq y$, say $x \triangleright y$ (“ x reads y ”) iff $\Pi_x \nabla_x E$ depends on the y -block somewhere on $\mathcal{X}^\circ$. Let Γ be the graph on $\mathcal{V}$ with edge $\{x, y\}$ iff $x \triangleright y$.

Lemma 3.3 (No one-way coupling). *For $x \neq y$ the following are equivalent:*

- (i) $x \triangleright y$;
- (ii) $\exists u \in T_x, v \in T_y, Q \in \mathcal{X}^\circ$ with $u^\top \nabla^2 E(Q) v \neq 0$;
- (iii) $y \triangleright x$.

In particular $\triangleright$ is symmetric and Γ is a well-defined undirected graph.

Proof. (ii) $\Rightarrow$ (i). Let $v \in T_y$ and $|s|$ small, so that $Q + sv \in \mathcal{X}^\circ$ (one block moves inside its open simplex; the product structure of $\mathcal{X}^\circ$ makes this legitimate). Put $g(s) = u^\top \Pi_x \nabla_x E(Q + sv) = u^\top \nabla_x E(Q + sv)$, using $\Pi_x^\top u = u$. Then $g'(0) = u^\top \nabla^2 E(Q) v \neq 0$, so g is non-constant on the segment, whose endpoints differ only in the y -block. Hence $x \triangleright y$.

(i) $\Rightarrow$ (ii), *by contraposition*. Suppose $u^\top \nabla^2 E v = 0$ for all $u \in T_x, v \in T_y$, everywhere on $\mathcal{X}^\circ$. Take $P, P' \in \mathcal{X}^\circ$ differing only in the y -block and set $v := P' - P \in T_y$; the segment $[P, P']$ lies in $\mathcal{X}^\circ$ by convexity of Δ_y° and the product structure. Let $\varphi(s) = \Pi_x \nabla_x E(P + sv) \in T_x$. Then $\varphi'(s) = \Pi_x \nabla^2 E(P + sv) v$ and, for every $u \in T_x, u^\top \varphi'(s) = u^\top \nabla^2 E v = 0$, so $\varphi'(s) \perp T_x$. But φ is T_x -valued, hence so is φ' (a limit of difference quotients of T_x -valued maps; T_x is closed). Therefore $\varphi' \in T_x \cap T_x^\perp = \{0\}$, so φ is constant and $\Pi_x \nabla_x E(P) = \Pi_x \nabla_x E(P')$. As P, P' were arbitrary, $\neg(x \triangleright y)$.

(ii) $\Leftrightarrow$ (iii). Condition (ii) is symmetric under $(x, u) \leftrightarrow (y, v)$ because $\nabla^2 E(Q)$ is a symmetric matrix at every Q . This is Schwarz’s theorem, and it is exactly where Assumption 1 is used. $\square$

Remark 3.4 (Gauge invariance; the shift loophole, closed structurally). Condition (ii) involves only derivatives along directions in $\prod_x T_x$, i.e. only the restriction $E|_{\mathcal{X}}$: if E and E' agree on $\mathcal{X}$, then $E - E'$ vanishes on the relatively open subset $\mathcal{X}^\circ$ of the affine hull, so all its tangential derivatives vanish there and $\Gamma(E) = \Gamma(E')$.

A concrete instance of the loophole: the term $c f(Q_y) \sum_s Q_x(s)$ has non-zero *ambient* mixed partials, but for $u \in T_x$ we get $u^\top \nabla_x[\cdot] = c f(Q_y) \mathbf{1}^\top u = 0$; and its contribution to $\nabla_y E$ is $c \nabla f(Q_y) (\mathbf{1}^\top Q_x)$, whose x -dependence dies tangentially in the same way. Both directions are dead — consistently with symmetry — and correctly so: restricted to $\mathcal{X}$ the term equals $c f(Q_y)$, which carries no information about Z_x . This is the general shape of the loophole: *a coupling invisible to x ’s softmax is precisely one that, on the feasible set, does not depend on x ; so nothing can be read through it in either direction.*

Note also that the naive ambient criterion $\partial^2 E / \partial Q_x(s) \partial Q_y(s') \neq 0$ defines a *supergraph* of Γ ; the tangential Γ is the sharp object, and it is what makes the loophole disappear by construction.

4 Influence is confined to Γ -components

For a node x write $\text{Comp}(x)$ for its Γ -connected component and $N(x)$ for its neighbour set.

Lemma 4.1 (Confinement; head variables handled explicitly). *Fix any schedule. For every $t \geq 0$ and every node x there is a function $\Phi_{x,t}$ with*

$$Q_x^{(t)} = \Phi_{x,t} \left((Q_y^{(0)})_{y \in \text{Comp}(x)}, (w_{\pi(y)})_{y \in \text{Comp}(x) \cap \{\text{label nodes}\}} \right)$$

for all interior initialisations and all words. That is, information reaches $Q_x^{(t)}$ only from initial blocks in x 's component and from words whose entry node z_k lies in that component.

Proof. Induction on t . For $t = 0$, $Q_x^{(0)}$ is its own initialisation; for $x = z_k$ it is $\iota_k(w_k)$ — own block, own word — and $x \in \text{Comp}(x)$.

For the step: nodes outside B_t carry over and the claim holds by the inductive hypothesis. For $x \in B_t$,

$$Q_x^{(t)} = \text{softmax} \left(-\frac{1}{\lambda_{x,t}} \nabla_x E(Q^{(t-1)}) \right),$$

which by Lemma 3.1 is a function of $\Pi_x \nabla_x E(Q^{(t-1)})$ and, if x is a label node, of $w_{\pi(x)}$ through the unary $u_{\pi(x)}$ of Assumption 3 — and of no other word, again by Assumption 3. By Lemma 3.3 (direction (i), applied blockwise) $\Pi_x \nabla_x E$ does not depend on any block $y \notin N(x) \cup \{x\}$; iterating over the finitely many blocks, each single-block move staying inside $\mathcal{X}^\circ$, shows it is a function of $(Q_y^{(t-1)})_{y \in N(x) \cup \{x\}}$ alone. Each such $Q_y^{(t-1)}$ is, by the inductive hypothesis, a function of the initialisations and words of $\text{Comp}(y) = \text{Comp}(x)$. Composing gives the claim.

Head variables receive no special treatment, and need none: any route “through an H ” consists of Γ -edges incident to that head node and is confined exactly like every other route. $\square$

Remark 4.2 (What is *not* claimed). Lemma 4.1 is the necessity direction only: *no path* $\Rightarrow$ *no influence*. We do *not* claim “path $\Rightarrow$ influence”. That is a faithfulness statement, false in general (couplings can cancel, or sit dead on the reachable set), and Proposition 8.1 exhibits exactly this phenomenon. No step below needs the sufficiency direction. This is the honest reading of “multi-step influence propagates along Γ -paths”: it is a statement of *confinement*, not of transmission.

5 Causality kills both directions

Proposition 5.1 (C2 $\iff$ no cross-position edges). *Assume SE. Then C2 holds if and only if Γ has no edge between nodes of different positions — equivalently, every Γ -component is contained in a single position's node set.*

Proof. ($\Rightarrow$) Let x, y with $\pi(x) < \pi(y)$. Apply C2 to the schedule $B_1 = \{x\}$ at $t = 1$: for all $Q^{(0)} \in \mathcal{X}^\circ$, $\text{softmax}(-\frac{1}{\lambda} \nabla_x E(Q^{(0)}))$ does not depend on the y -block. Because C2 quantifies over *all* interior initialisations, this is an identical statement on $\mathcal{X}^\circ$; by Lemma 3.1 it is equivalent to $\neg(x \triangleright y)$. By Lemma 3.3, also $\neg(y \triangleright x)$.

This is the crux of the theorem: *Schwarz symmetry converts “the later node's block does not feed the earlier node's update” into “the earlier node's block does not feed the later node's update”*. The pair was an arbitrary cross-position pair, so no edge joins different positions.

($\Leftarrow$) If components are position-pure, Lemma 4.1 gives that every $Q_x^{(t)}$, under every schedule, is a function of position- $\pi(x)$ initial blocks and $w_{\pi(x)}$ only, which implies C2. $\square$

Theorem 5.2 (Main impossibility). *Let a scheme satisfy SE (with Assumptions 1–4) and C2. Then for every schedule, every t and every i ,*

$$Q_{z_i}^{(t)} = \Phi_{i,t}(w_i, \text{position-}i \text{ initial blocks}).$$

*In particular $\neg X$: no label marginal ever depends on any other word. Equivalently, **SE, C2 and X cannot hold simultaneously**. Contrapositively: any update scheme that is network-causal and contextual is not generated, in the sense of Assumption 2, by any single C^2 word-local energy.*

Proof. SE and C2 give position-pure components by Proposition 5.1. Word entry is confined to label nodes by Assumptions 3–4, so Lemma 4.1 yields the displayed locality, whence $\neg X$. $\square$

Corollary 5.3 (Word-level causality follows, vacuously). $C2 \Rightarrow C1$. *Indeed, word initialisations are particular interior initialisations, and by Theorem 5.2 $Q_{z_i}^{(t)}$ depends on no other word at all — future or past.*

This is the sting of the theorem: under a single energy, the only way to be causal is to be trivial.

6 Pairwise compatibility: the three witnesses

C2 $\wedge$ SE, $\neg$ X. Take $E = \sum_i u_i(Q_i; w_i) + \sum_{i,c} \psi_{ic}(Q_{z_i}, Q_{h_{ic}})$: all couplings are within-position. Then Γ has no cross-position edge, so C2 holds by Proposition 5.1; SE holds by construction; $\neg$ X by Theorem 5.2.

SE $\wedge$ X, $\neg$ C. The original PT. It has a single energy (Appendix A of the paper), is contextual for generic $\mathbf{T}$, and is not causal: term 2 of Eq. (4) is precisely a later block feeding an earlier update. Equivalently, the trilinear term $Q_{jc}(i)Q_j(b)Q_i(a)\mathbf{T}_{b,a}^{(c)}$ creates a cross-position edge, which by Lemma 3.3 fires in both directions.

C2 $\wedge$ X, $\neg$ SE. Approach A: mask the heads ($\mathcal{D}_i = \{1, \dots, i-1\}$) and keep only term 1 in the label message. C2 holds for *all* activations: the h_{ic} -update reads $\{Q_{z_i}\} \cup \{Q_{z_j} : j < i\}$, and the z_i -update reads $\{Q_{h_{ic}}\}_c \cup \{Q_{z_j} : j < i\}$, so every activation at position i is a function of blocks at positions $\leq i$, for arbitrary interior states. X holds for generic $\mathbf{T}$. $\neg$ SE is forced by Theorem 5.2, and is verified directly — with the exact degeneracy characterised — in Corollary 7.1.

All three pairs are therefore realisable; the triple is not.

7 Approach A is not a gradient field: the exact statement

Corollary 7.1. *Suppose some E satisfying Assumptions 1 and 3 reproduces Approach A’s updates in the sense of Assumption 2, for some fixed $\lambda_Z, \lambda_H > 0$. Then every channel matrix is tangentially degenerate:*

$$\Pi \mathbf{T}^{(c)} \Pi = 0, \quad i.e. \quad \mathbf{T}_{a,b}^{(c)} = \alpha_a^{(c)} + \beta_b^{(c)}$$

for some vectors $\alpha^{(c)}, \beta^{(c)}$. In that case the tangential label message of Approach A does not depend on the state, so the scheme is non-contextual anyway. Hence: non-degenerate $\mathbf{T} \Rightarrow$ no such E exists; degenerate $\mathbf{T} \Rightarrow \neg X$. There is no third option.

Proof. Approach A’s label field (the vector inside the softmax) is

$$m_{z_i}(a) = \mathbf{S}_{w_i,a} + \sum_c \sum_{j < i} \sum_b Q_{ic}(j) Q_j(b) \mathbf{T}_{a,b}^{(c)}.$$

If E reproduces the updates then, by Lemma 3.1, $\Pi_{z_i} \nabla_{z_i} E = -\lambda_Z \Pi_{z_i} m_{z_i}$ on $\mathcal{X}^\circ$; state-dependent shifts cannot rescue anything, since for tangential u we have $u^\top \mathbf{1} (\nabla c)^\top v = 0$.

Take $i > j$ and tangential u, v , and compute the same tangential Hessian entry in two ways:

$$u^\top \nabla^2 E v \stackrel{\text{via } z_i\text{-field}}{=} -\lambda_Z \sum_c Q_{ic}(j) u^\top \mathbf{T}^{(c)} v, \quad u^\top \nabla^2 E v \stackrel{\text{via } z_j\text{-field}}{=} 0,$$

the second because m_{z_j} sums only over $k < j$ and therefore contains no Q_{z_i} . Schwarz forces equality at every $Q \in \mathcal{X}^\circ$. The coordinates $Q_{ic}(j)$, $c = 1, \dots, h$, belong to h independent

blocks and each ranges freely over $(0, 1)$ as Q ranges over $\mathcal{X}^\circ$; varying one block at a time gives $u^\top \mathbf{T}^{(c)} v = 0$ for every c and all tangential u, v , i.e. $\Pi \mathbf{T}^{(c)} \Pi = 0$, i.e. $\mathbf{T}_{a,b}^{(c)} = \alpha_a + \beta_b$.

With such $\mathbf{T}$ the head logits become $F_{ic}(j) = \langle Q_i, \alpha \rangle + \langle Q_j, \beta \rangle$, and the label message contribution is $\sum_j Q_{ic}(j) \sum_b Q_j(b)(\alpha_a + \beta_b) = \alpha_a + (\text{a shift in } a)$, whose centred part is the constant $\Pi\alpha$, independent of the state. No context reaches any label marginal. $\square$

Remark 7.2 (Consistency check). The *unmasked* PT field does pass the symmetry test, as it must, being an actual gradient:

$$\frac{\partial m_{z_i}}{\partial Q_{z_j}} = \sum_c \left[Q_{ic}(j) \mathbf{T}^{(c)} + Q_{jc}(i) \mathbf{T}^{(c)\top} \right],$$

and the required identity $u^\top (\partial m_{z_i} / \partial Q_{z_j}) v = v^\top (\partial m_{z_j} / \partial Q_{z_i}) u$ holds term by term under the swap $(i, u) \leftrightarrow (j, v)$ with transposition. *Term 1 of i and term 2 of j are the two faces of one Hessian entry.* Masking discards one face; Schwarz says one cannot.

8 Sharpness: the trajectory version of the theorem is false

Proposition 8.1. *There is a scheme satisfying SE, C1 and X simultaneously. Hence causality must be formalised as C2 (or C1 must be supplemented by a faithfulness hypothesis) for the impossibility to hold.*

Proof. Take $n = 2$ positions, binary labels ($d = 2$), and no head variables (permitted: the factor graph simply has none, or has inert position-local ones). Let $\mathcal{W}$ be the finite vocabulary and assume the embeddings $\iota(v) = \text{softmax}(\mathbf{S}_{v,\cdot})$ are pairwise distinct with pairwise distinct gaps $\Delta(v) := \iota(v)_1 - \iota(v)_2$ (generic $\mathbf{S}$). Put $\lambda = 1$ and

$$\begin{aligned} E(Q; w) &= -\langle \mathbf{S}_{w_1,\cdot}, Q_1 \rangle - \langle \mathbf{S}_{w_2,\cdot}, Q_2 \rangle + \psi(Q_1) \varphi(Q_2), \\ \psi(Q_1) &= h(Q_1(1) - Q_1(2)), \quad \varphi(Q_2) = Q_2(1), \end{aligned}$$

where h is the univariate Hermite interpolation polynomial with

$$h(\Delta(v)) = c_v \quad (\text{pairwise distinct}), \quad h'(\Delta(v)) = 0 \quad \text{for every } v \in \mathcal{W}$$

(degree at most $2|\mathcal{W}| - 1$; existence is standard). Then E is polynomial, hence C^2 , and word-local (the pair term is word-free): SE holds.

C1. We claim $Q_1^{(t)} = \iota(w_1)$ for every schedule and every t . Induction: $Q_1^{(0)} = \iota(w_1)$. Whenever z_1 is updated, the tangential part of its gradient is $\Pi[-\mathbf{S}_{w_1,\cdot} + \varphi(Q_2) \nabla_{Q_1} \psi]$, and along the tangent direction $(1, -1)$ the term $\nabla \psi$ contributes $2h'(Q_1(1) - Q_1(2)) = 2h'(\Delta(w_1)) = 0$ at the current state. Hence the update returns $\text{softmax}(\mathbf{S}_{w_1,\cdot}) = \iota(w_1)$: the reachable set of z_1 is the fixed point $\{\iota(w_1)\}$, closed under every schedule (synchronous included), and $Q_1^{(t)}$ never depends on w_2 or on Q_2 . So C1 holds.

X. Under the schedule $B_1 = \{z_2\}$, the tangential gradient of z_2 at $t = 1$ contains the term

$$-\psi(Q_1^{(0)}) \Pi(1, 0)^\top = -c_{w_1} \Pi(1, 0)^\top,$$

and the values c_{w_1} are pairwise distinct, so $Q_2^{(1)}$ varies with w_1 : the scheme is contextual (past $\rightarrow$ future, exactly the useful direction).

C2 fails, as Theorem 5.2 requires: at states where $Q_1(1) - Q_1(2)$ lies off the finite critical set we have $h' \neq 0$ (otherwise h is constant, contradicting distinctness of the c_v), and the cross-position edge fires. The backward channel exists in E ; it is merely never *reached*. $\square$

Remark 8.2 (Anatomy of the gap). C1 constrains the scheme on the reachable set generated by a finite vocabulary — finitely many algebraic conditions per horizon. C2 (and the graph condition) are identical-vanishing conditions on $\mathcal{X}^\circ$. Proposition 8.1 lives exactly in the daylight between them: the acausal coupling’s coefficient is a polynomial tuned to vanish on the reachable set. Three honest observations:

- (i) It is a *parameter conspiracy*, not an architecture: h is fitted to the specific rows of $\mathbf{S}$, and a generic perturbation of $\mathbf{S}$ (with h fixed) breaks C1.
- (ii) It does not survive richness: with an infinite initialisation family accumulating in Δ° and real-analytic E , the identity theorem forces $h' \equiv 0$ on an interval, collapsing the construction — for *direct* label–label edges. Multi-step head-mediated routes would need a separate argument, which is not claimed here.
- (iii) Within the strictly multi-affine (factor-graph) class over binary labels, a direct pair coupling cannot play this trick — ψ affine makes $d\psi$ constant, and one interior zero kills both directions — but head-mediated coefficients reopen the question, which is why the clean unconditional theorem is the C2 one.

If a C1-based statement is wanted for a paper, the defensible phrasing is: *C1 together with faithfulness (no exact cancellation on the reachable set) implies the graph condition, hence $\neg X$; and faithfulness holds off a proper algebraic subvariety of parameter space for factor-graph energies* — with the last clause flagged as asserted-generic rather than proven.

9 Loopholes, closed with the mechanism named

(a) Softmax shift-invariance.

Closed structurally rather than by patching: coupling is *defined* tangentially (Section 3), which is exactly the quotient by shifts (Lemma 3.1); Remark 3.4 proves that the invisible couplings are precisely those carrying no information about the target variable on the feasible set, and that Γ depends only on $E|_{\mathcal{X}}$. Nothing exploitable exists on this axis, in either direction.

(b) Message weights λ .

Each $\lambda_{x,t} > 0$ acts on the centred gradient as a positive scalar, so Lemma 3.1’s injectivity and all zero patterns are unchanged. Per-node and per-step variation (extended entropic Frank–Wolfe; annealing) is already covered by Assumption 2. Distinct λ ’s cannot manufacture asymmetry: in Corollary 7.1 they multiply through and cancel against the zero on the other side.

(c) Schedules.

Causality and confinement were proven *for every* schedule: Lemma 4.1 inducts over an arbitrary schedule, and Proposition 5.1 uses only the freedom to schedule a single node once, which any schedule notion includes. Asynchrony composes the same Γ -edges; it cannot create them.

(d) Higher-order and non-multilinear potentials.

Multilinearity was never used — only Assumption 1. Arbitrary factor arities and arbitrary smooth non-factor-graph energies are covered. (Multi-affinity appears only in Remark 8.2(iii), as a *restriction* that makes the counterexample harder to build.)

(e) Update-rule variants.

Damped and step-size variants (Appendices B.1–B.2 of the paper): if the update is $Q_x^{(t)} =$

$\alpha \text{softmax}(-g/\lambda) + (1 - \alpha) Q_x^{(t-1)}$, or messages are mixed with memory, the per-step map is still, for fixed local state, an injective function of the centred gradient, and Proposition 5.1’s extraction and Lemma 4.1’s confinement go through verbatim (the auxiliary memory is position-local state). In general the theorem holds for any update family in which each node’s step is an injective function of $(Q_x^{\text{old}}, \Pi_x \nabla_x E(Q))$. Injectivity is where honesty lives: an update that *ignores part of* the centred gradient is precisely a hand-deletion — Approach A — and then SE fails by definition, which is the theorem’s conclusion arriving early. Final-iteration unnormalised log-space outputs (as PT uses downstream) read the same centred gradient affinely; nothing changes.

(f) Scope.

The mean-field family here is fully factorised with independent simplex blocks (PT’s setting: head selection, no tree constraint). Projected updates onto *coupled* constraint sets (e.g. arborescence polytopes) change the update operator and lie outside this theorem’s scope; a different theorem would be needed there. Also, C2 is the all-activations notion; a label-only variant reopens mild faithfulness issues for two-step head-mediated composites (a head node could read the future while the label node reads the head node only on a flat region). Since masked attention and Approach A satisfy the all-activations version anyway — attention maps in a real decoder are themselves causal — nothing of practical relevance is lost by using C2.

10 What Part I establishes

1. No redefinition of ϕ_t , and no new potential structure, can make causality a structural property *inside a single energy* while keeping the model contextual (Theorem 5.2). The obstruction is Schwarz’s theorem, not a defect of the particular potentials.
2. In particular, the single trilinear term $Q_{jc}(i)Q_j(b)Q_i(a)\mathbf{T}_{b,a}^{(c)}$ has two faces: differentiated by $Q_{jc}(i)$ it is “word j may attend to word i ” — the message that makes causal attention possible at all — and differentiated by $Q_i(a)$ it is term 2, the leak. One term, both roles.
3. Approach A is therefore not repairable by a cleverer energy: either $\mathbf{T}$ is tangentially degenerate and the model is non-contextual, or no energy exists (Corollary 7.1).
4. Consequently the model class must change. Part II carries this out.

Part II — The directed reformulation

11 An instructive failure: the structural backward-arc restriction

Before the construction that works, one proposal that does not — worked out honestly, because it isolates exactly which check fails and why.

Proposal A’. Restrict the *domain* of the head variables: $H_i^{(c)} \in \{1, \dots, i-1\}$ (and $\{\text{ROOT}\}$ for $i = 1$). The potential is unchanged in form; arcs simply cannot point forward. This is a legitimate CRF — the mask is structural, not a hand-edit of an update.

Energy.

$$E(Q) = - \sum_i \sum_a Q_i(a) \mathbf{s}_{w_i, a} - \sum_c \sum_i \sum_{j < i} \sum_{a, b} Q_{ic}(j) Q_i(a) Q_j(b) \mathbf{T}_{a, b}^{(c)}.$$

Gradients.

$$\begin{aligned} \frac{\partial E}{\partial Q_k(a)} &= -\mathbf{s}_{w_k, a} - \underbrace{\sum_c \sum_{j < k} \sum_b Q_{kc}(j) Q_j(b) \mathbf{T}_{a, b}^{(c)}}_{\text{term 1: past only}} - \underbrace{\sum_c \sum_{i > k} \sum_b Q_{ic}(k) Q_i(b) \mathbf{T}_{b, a}^{(c)}}_{\text{term 2: } i > k, \text{ the future}} \\ \frac{\partial E}{\partial Q_{kc}(j)} &= - \sum_{a, b} Q_k(a) Q_j(b) \mathbf{T}_{a, b}^{(c)}, \quad j < k. \end{aligned}$$

Verification. Check 2 (energy exists): **passes** — this is an honest gradient of a real energy. Check 3 (valid distribution): **passes** — softmax. Check 1 (causality): **fails** — the third sum contains Q_i for $i > k$. Restricting arcs to point backward does not stop information flowing forward-to-backward *through* the factor. Check 4: it reduces to masked attention *plus* a transpose-attention term (the $Q_{h, c}^\top$ -weighted summand of the unsimplified Eq. (16), with $Q_{h, c}$ triangular).

Reading. A' is exactly complementary to Approach A: A' keeps the energy and loses causality; A keeps causality and loses the energy. Together they *instantiate* Theorem 5.2. Note also the correct probabilistic reading of A' : Q_k is the *smoothing* marginal of a directed-arc CRF, and future words attending to k is legitimate evidence about Z_k . That is fine for an encoder and wrong for generation — which is the first hint that the fix is a change of inference query, not of potentials.

12 Proposal B: a directed chain of conditional CRFs

12.1 Model class

The change of class is from one globally normalised undirected CRF to a *Bayesian network whose conditionals are small CRFs* — a locally normalised directed model, which is what an autoregressive decoder is anyway. For each position i define a conditional CRF over the block $(Z_i, \{H_i^{(c)}\}_c)$ given the realised prefix labels $z_{< i}$:

$$p(Z_i, \{H_i^{(c)}\} \mid z_{< i}, w_{1:i}) = \frac{1}{Z_i(z_{< i})} \exp \left(\mathbf{s}_{w_i, Z_i} + \sum_c \sum_{j \in \mathcal{D}_i} \mathbf{1}[H_i^{(c)} = j] \tau_{Z_i}^{(c)}(j; z_j) \right),$$

with head domain $\mathcal{D}_i = \{\text{ROOT}, 1, \dots, i-1\}$, arc score $\tau_a^{(c)}(j; z_j) = \mathbf{T}_{a, z_j}^{(c)}$ for $j < i$, and $\tau_a^{(c)}(\text{ROOT}) = r_a^{(c)}$ a learned root/sink column. The root column makes $i = 1$ well-posed: the first word attends to ROOT with probability one, so the *attention sink* falls out of the model rather than being bolted on. The joint over the sentence is the chain-rule product

$$p(z, h \mid w) = \prod_{i=1}^n p(z_i, h_i \mid z_{< i}, w_{1:i}),$$

a valid probability model by construction.

Remark 12.1 (Why the acausal term is not born). In a directed model, the factor in which position i serves as the *head of a later word* j lives inside $p(Z_j, H_j \mid z_{<j})$ — a conditional in which z_i is a *conditioning value*, not an inference variable. Local normalisation $\mathcal{Z}_j(z_{<j})$ runs over (Z_j, H_j) only, so no backward coupling arises from it either. This is the structural reason, and it is the reason Theorem 5.2 is respected rather than contradicted: there is no single global energy here, and by Theorem 5.2 there cannot be.

12.2 Inference: variational filtering

Process $i = 1, \dots, n$, maintaining frozen filtering marginals $\bar{q}_j(\cdot) \approx p(Z_j \mid w_{1:j})$ for $j < i$. Because the log-potential is *linear in the one-hot encoding of z_j* , the expected log-conditional under $\bar{q}$ is exact at the potential level:

$$\mathbb{E}_{\bar{q}_j}[\mathbf{T}_{a,Z_j}^{(c)}] = \sum_b \bar{q}_j(b) \mathbf{T}_{a,b}^{(c)} =: B_{j,a}^{(c)}.$$

This is precisely the manoeuvre already proved correct in Appendix B.3.1 of the paper: a neighbour variable with a fixed distribution collapses into a binary factor with contracted scores B . There the fixed-distribution neighbours were the observed global feature variables F_j ; here they are the frozen prefix marginals $\bar{q}_j$. Step i is therefore exact MFVI on a small CRF over $(Z_i, \{H_i^{(c)}\})$ with unary $\mathbf{S}_{w_i,\cdot}$ and binary factors $\phi_b(H_i^{(c)}=j, Z_i=a) = \exp(B_{j,a}^{(c)})$.

Per-step energy. Variables: $Q_i, \{Q_{ic}\}$. Constants: $\mathbf{S}_{w_i,\cdot}$ and $\{B_{j,\cdot}^{(c)}\}_{j \in \mathcal{D}_i}$.

$$E_i(Q_i, \{Q_{ic}\}) = - \sum_a Q_i(a) \mathbf{S}_{w_i,a} - \sum_c \sum_{j \in \mathcal{D}_i} \sum_a Q_{ic}(j) Q_i(a) B_{j,a}^{(c)}$$

Gradients — nothing else appears.

$$\frac{\partial E_i}{\partial Q_i(a)} = -\mathbf{S}_{w_i,a} - \sum_c \sum_{j \in \mathcal{D}_i} Q_{ic}(j) B_{j,a}^{(c)}, \quad \frac{\partial E_i}{\partial Q_{ic}(j)} = - \sum_a Q_i(a) B_{j,a}^{(c)}.$$

Updates (with the paper’s message weights; the asynchronous variant of Section 2.3.2 applies verbatim):

$$Q_{ic}(j) \propto \exp\left(\frac{1}{\lambda_H} \sum_a Q_i(a) B_{j,a}^{(c)}\right), \quad j \in \mathcal{D}_i; \quad Q_i(a) \propto \exp\left(\frac{1}{\lambda_Z} \left[\mathbf{S}_{w_i,a} + \sum_c \sum_j Q_{ic}(j) B_{j,a}^{(c)}\right]\right).$$

Iterate the inner loop to convergence (or for a fixed count), then freeze $\bar{q}_i := Q_i^{\text{final}}$ and advance to $i + 1$.

12.3 The four checks

Check 1 (causality): passes, by induction. Claim C_i : $\bar{q}_i$ and all inner iterates at position i are functions of $w_{1:i}$ only. Base: the constants of E_1 are $\mathbf{S}_{w_1,\cdot}$ and $r^{(c)}$ (a parameter), so C_1 . Step: the constants of E_i are $\mathbf{S}_{w_i,\cdot}$ and $B_{j,\cdot}^{(c)} = f(\bar{q}_j)$ for $j < i$; by the inductive hypothesis each $\bar{q}_j$ depends on $w_{1:j} \subseteq w_{1:i-1}$; every inner iterate is a function of the constants of E_i . Hence C_i . No Q_j with $j > i$ appears anywhere: such quantities are not variables of E_i , and no constant of E_i contains them.

Check 2 (energy exists): passes, with the caveat stated precisely. Each update is exactly $\text{softmax}(-\nabla E_i/\lambda)$; nothing is deleted. Term 2 does not arise because the factors in which i heads a future j belong to E_j , where differentiation is with respect to Q_j, Q_{jc} only.

What must be said plainly: *there is no single global energy generating all updates* — and by Theorem 5.2 *there cannot be*. The global object that exists instead is the valid directed joint $\prod_i p(z_i, h_i \mid z_{<i}, w_{1:i})$, and the inference is variational *filtering* (assumed-density style): $Q_i \approx p(z_i \mid w_{1:i})$, not the smoothing marginal $p(z_i \mid w_{1:n})$. The remaining approximation is the one MFVI already makes — replacing the true prefix posterior by a product of marginals; the expectation-inside-log step is exact here by log-linearity, as shown above. For a decoder, filtering is the semantically correct object, so the white-box interpretation is not merely preserved: it now matches the generative use case.

One clarification that pre-empts a standard confusion: *frozen* means constant with respect to step- i 's inference problem, *not* detached in autodiff. Training gradients flow backward through $B_{j,\cdot}^{(c)}$ into $\bar{q}_j$ exactly as they flow through cached activations in a causal transformer. Forward causality is what defines a decoder; backward gradient flow is what trains it.

Check 3 (valid distribution): passes. Softmax over labels; Q_{ic} normalises over $\mathcal{D}_i$ including ROOT.

Check 4 (attention correspondence): passes, and more cleanly than in the encoder. Decompose $\mathbf{T}^{(c)} = \mathbf{U}^{(c)}\mathbf{V}^{(c)\top}$ (Eqs. 14 and 21 of the paper). Then

$$e_{ij}^{(c)} = \frac{1}{\lambda_H} (Q_i \mathbf{U}^{(c)}) (\bar{q}_j \mathbf{V}^{(c)})^\top, \quad j < i; \quad G_i = \sum_c \left[\sum_{j \in \mathcal{D}_i} \alpha_{ij}^{(c)} (\bar{q}_j \mathbf{V}^{(c)}) \right] \mathbf{U}^{(c)\top},$$

$$Q_i = \sigma \left(\frac{\mathbf{S}_{w_{i,\cdot}} + G_i}{\lambda_Z} \right).$$

This is masked scaled dot-product attention verbatim: the query $Q_i \mathbf{U}^{(c)}$ comes from the current token's evolving state, and keys = values = $\bar{q}_j \mathbf{V}^{(c)}$ come from *frozen* prefix states — so **the KV cache falls out of the model** rather than being an engineering trick. Three structural bonuses relative to Section 3 of the paper:

- (a) the factor 2 of Eq. (20) disappears, since no symmetrisation is performed;
- (b) *no symmetry assumption on $\mathbf{T}^{(c)}$ is needed* — head-child order is now genuinely meaningful, so the causal PT is a *directed* dependency model, whereas the encoder's correspondence had to assume undirected, link-grammar-style symmetry;
- (c) the excluded diagonal ($j \neq i$) is unproblematic, because the re-injected unary $\mathbf{S}_{w_{i,\cdot}}$ at each iteration plays the residual role, exactly as in the paper.

The distance variant (Eq. 9) carries over with only the $i - j > 0$ half of the RPE table — halving those parameters.

Scheduling. Two options, both causal. *Serial left-to-right* with inner convergence gives the clean filtering story: each E_i is a well-posed two-block MFVI problem inheriting the per-step entropic Frank–Wolfe guarantees. *Layer-parallel*: iteration t computes all $Q_i^{(t)}$ from $\{Q_j^{(t-1)}\}_{j < i}$ under a strict lower-triangular mask; paths still run only from the past, and the computation graph is *exactly* that of a depth- T , parameter-shared causal transformer.

13 Proposal C: per-prefix re-inference

The other principled exit from Theorem 5.2: keep the original undirected PT, but define the causal representation of position i as the output of running the *encoder on the prefix* $w_{1:i}$ and exporting the last position:

$$Q_i := Q_i^{[i]}, \quad Q^{[i]} = \text{MFVI fixed point of } E^{[i]},$$

where $E^{[i]}$ is the original PT energy (Eqs. 1–8, arcs unrestricted) built on words $1, \dots, i$ only. The gradients are identical to Eq. (4) with all sums truncated at i ; for the exported variable $k = i$,

$$\frac{\partial E^{[i]}}{\partial Q_i(a)} = -\mathbf{s}_{w_i, a} - \sum_c \sum_{j < i} \sum_b \left(Q_{ic}^{[i]}(j) Q_j^{[i]}(b) \mathbf{T}_{a,b}^{(c)} + Q_{jc}^{[i]}(i) Q_j^{[i]}(b) \mathbf{T}_{b,a}^{(c)} \right).$$

Note that **term 2 survives — legally**: here $j < i$, so “past words choosing i as head” is within-prefix evidence, which Proposal B’s frozen past cannot deliver.

Checks. Check 1: passes — $E^{[i]}$ contains no symbol from $w_{>i}$, so every $Q_k^{[i]}$ is a function of $w_{1:i}$ and the export is causal. Check 2: passes per prefix — each export is honest MFVI of a real energy $E^{[i]}$; but again a *different* subproblem per position. A single global E is excluded by Theorem 5.2, and here the family $\{E^{[i]}\}$ plays the role that the directed factorisation plays in B. (CRFs are per-input models anyway, so a per-prefix family is not exotic.) Check 3: passes. Check 4: each position’s output is full *bidirectional* attention over its prefix — a prefix-LM-style column, not standard causal attention. Representations of earlier positions are recomputed and change per prefix (only the diagonal is exported), so there is no KV cache and the cost is n encoder passes per sequence.

B versus C, in one line. B is filtering with a frozen past (transformer decoder; cacheable; term 2 absent even where it would be legal). C is smoothing within the prefix (richer per-position posterior, including legal backward evidence; factor- n compute; no incremental consistency of the prefix states). Both are corollaries of Theorem 5.2: causality forces per-position subproblems, and B and C are the two natural choices of subproblem.

14 Variants that do not escape the theorem

- **Asymmetric or triangular $\mathbf{T}$** (in label space), or separate forward/backward score matrices: these still place one factor on (Z_i, Z_j) , so Lemma 3.3 applies. They change coefficients, not the dependency pattern.
- **Clamping future marginals to uniform inside the joint E** : the residual term-2 field becomes a length-dependent constant (it leaks n , and is undefined during generation), and the updates are slice-restricted gradients of different slices per position — strictly worse than B, with muddier semantics.
- **Writer/reader copies of each Z_i tied by an equality factor**: the equality factor is itself a symmetric coupling, so Lemma 3.3 applies. Enforcing the copy by *substitution of a computed value* instead of by a factor is exactly the $B_{j,a}^{(c)}$ construction — i.e. it collapses into Proposal B.

Recommendation. Implement B. It is the honest decoder: a valid directed probabilistic model, exact per-step energies, an exact masked-attention reduction, a KV cache for free, and it drops the encoder’s symmetry assumption — directed dependencies for a directed task. Keep C as the conceptual gold standard and as an ablation (does legal within-prefix term 2 buy anything?). Use Part I verbatim in the “why not simpler fixes” section of the write-up: it upgrades “naive masking is insufficient” from an observation to a theorem, and it treats the undirected-versus-directed tension head-on.

Part III — The output mechanism

15 The problem, and why the construction is nearly forced

The rejected design was: take $Q_i^{(T)}$ — the posterior over the *label* of word i — multiply by a learned $d \times |\mathcal{V}|$ matrix, softmax, call the result the distribution over word $i + 1$, and train with shifted targets. The objection is correct and is threefold: that projection corresponds to *no factor* in the graph; Q_i is the posterior over the label of an *already-known* word, conditioned on w_i through its own unary, so it answers “what is the syntactic category of this known word”, not “what comes next”; and the arc factor connects the label of a word to the label of its *head*, never to the label of the next position, so no path in the graph leads from Z_i to the identity of w_{i+1} .

The repair is strongly constrained by the graph itself. The only factor touching a word variable is the word–label factor; the only factors touching a label are that factor and its arc factors. Hence *any* graph-faithful route from prefix to vocabulary must be

$$\text{prefix marginals} \xrightarrow{\text{arc factors}} \text{label of the unknown slot} \xrightarrow{\text{word–label factor}} \text{word}.$$

That is the construction below.

16 The model: the chain becomes generative over words

The chain of Part II is $p(z, h \mid w)$ — conditioned on all words. A model that *predicts* words must be a joint over words. Promote each word to a variable W_t :

$$p(w, z, h) = \prod_{t=1}^n p(W_t, Z_t, \{H_t^{(c)}\} \mid z_{<t})$$

with each step conditional the CRF

$$p(W_t, Z_t, H_t \mid z_{<t}) = \frac{1}{Z_t(z_{<t})} \exp \left(\underbrace{b_{W_t}}_{\text{word unary}} + \underbrace{\mathbf{S}_{W_t, Z_t}}_{\text{word–label factor}} + \sum_c \sum_{j \in \mathcal{D}_t} \mathbf{1}[H_t^{(c)} = j] \mathbf{T}_{Z_t, z_j}^{(c)} \right),$$

where $\mathcal{D}_t = \{\text{ROOT}, 1, \dots, t-1\}$, with the root column $B_{\text{ROOT}, a}^{(c)} = r_a^{(c)}$ as before. Three structural remarks, each load-bearing.

(a) The unary is not deleted; it is re-read. The unary $\phi_u(Z_t) = \exp(\mathbf{S}_{w_t, Z_t})$ was always the binary factor $\exp(\mathbf{S}_{W_t, Z_t})$ with W_t clamped to the observed word. Making W_t latent does not add a factor — it *unclamps* one. Observed mode and predictive mode are two conditioning patterns of one model, not two models. This is exactly the “word model” move of Wu’s thesis (§3.2), in which the word is a variable of the graph rather than an observed constant.

(b) Weight tying is forced, not chosen. The step conditional contains exactly one word-label factor. Whether one conditions on $W_t = w_t$ (encoding) or predicts W_t (decoding), the same matrix $\mathbf{S}$ mediates. Two different matrices for the two roles would mean two factors on the same pair, $\exp(\mathbf{S}_{w,a}) \exp(\mathbf{S}'_{w,a}) = \exp((\mathbf{S} + \mathbf{S}')_{w,a})$, which collapses back to one matrix. So input/output embedding tying — a heuristic in transformers, and one the PT paper itself applies to the MLM head — is here a *theorem* of graph-faithfulness, in the same family as the KV cache falling out of freezing.

(c) The word unary b_w is the bias of the LM head, as a factor. Optional but principled: a learned unigram potential on W_t . Set $b \equiv 0$ to drop it.

Remark 16.1 (What the chain rule now asserts). $W_t \perp w_{<t} \mid z_{<t}$: all predictive information flows through the label bottleneck, i.e. through the d -dimensional beliefs $\bar{q}_j$. This was already true of the representation in Part II; it is now load-bearing for prediction quality. It is the model's identity, not a defect — but it is the capacity constraint to watch (Section 20).

17 The predictive step: energy, updates, schedule

Fix slot t . The prefix is frozen as $\bar{q}_1, \dots, \bar{q}_{t-1}$ (filtering posteriors given $w_{1:t-1}$) and contracted as before into $B_{j,a}^{(c)} = \sum_b \bar{q}_j(b) \mathbf{T}_{a,b}^{(c)}$ — exact at the log-potential level by linearity in the one-hot encoding of z_j , unchanged from Part II. The live variables are $Q_W \in \Delta(\mathcal{V})$, $Q_Z \in \Delta([d])$ and $Q_c \in \Delta(\mathcal{D}_t)$.

Per-step energy.

$$E_t^\rightarrow(Q_W, Q_Z, \{Q_c\}) = - \sum_w Q_W(w) b_w - \sum_{w,a} Q_W(w) Q_Z(a) \mathbf{S}_{w,a} - \sum_c \sum_{j \in \mathcal{D}_t} \sum_a Q_c(j) Q_Z(a) B_{j,a}^{(c)}$$

Gradients — every term, nothing removed.

$$\begin{aligned} \frac{\partial E_t^\rightarrow}{\partial Q_W(w)} &= -b_w - \sum_a Q_Z(a) \mathbf{S}_{w,a}, & \frac{\partial E_t^\rightarrow}{\partial Q_c(j)} &= - \sum_a Q_Z(a) B_{j,a}^{(c)}, \\ \frac{\partial E_t^\rightarrow}{\partial Q_Z(a)} &= - \sum_w Q_W(w) \mathbf{S}_{w,a} - \sum_c \sum_{j \in \mathcal{D}_t} Q_c(j) B_{j,a}^{(c)}. \end{aligned}$$

Updates.

$$\begin{aligned} Q_c(j) &\propto \exp \left(\frac{1}{\lambda_H} \sum_a Q_Z(a) B_{j,a}^{(c)} \right), \\ Q_Z(a) &\propto \exp \left(\frac{1}{\lambda_Z} \left[\sum_w Q_W(w) \mathbf{S}_{w,a} + \sum_c \sum_j Q_c(j) B_{j,a}^{(c)} \right] \right), \\ Q_W(w) &\propto \exp \left(\frac{1}{\lambda_W} \left[b_w + \sum_a Q_Z(a) \mathbf{S}_{w,a} \right] \right). \end{aligned}$$

Observed mode is the same energy with Q_W clamped to δ_{w_t} : the word message $\sum_w Q_W(w) \mathbf{S}_{w,a}$ becomes $\mathbf{S}_{w_t,a}$, and $E_t^\rightarrow$ becomes literally the E_t of Part II. One energy family, two conditioning patterns.

17.1 Schedule and initialisation

Following the paper’s Eq. (7) rule — each variable initialises from its own unary-type factors — we set $Q_W^{(0)} \propto \exp(b)$, whence

$$Q_Z^{(0)} \propto \exp(\tilde{s}/\lambda_Z), \quad \tilde{s}_a = \sum_w Q_W^{(0)}(w) \mathbf{S}_{w,a},$$

a fixed learned vector: the prior-frequency-weighted mean embedding. This is the [MASK] token of MLM, or the learned query embedding of XLNet — *derived* rather than invented: the unknown word contributes its prior-average unary message.

Then run the asynchronous inner loop of Part II — update $\{Q_c\}$ (attention), then Q_Z — for τ rounds, and finish with a *single* Q_W update: **that softmax is the LM output layer**. Take $\tau \geq 2$ so that the attention query is context-dependent; at $\tau = 1$ the query is the fixed $\sigma(\tilde{s}/\lambda_Z)$, a learned probe — legal, but weaker. Iterating the $W \leftrightarrow Z$ loop further is legitimate MFVI, but it feeds the model’s own word guess back into the label belief (mean-field self-conditioning, mode-seeking); take the readout *before* feedback as the default and treat extra W -rounds as an ablation.

17.2 An exact-readout variant

The slot’s factor graph is a *star centred at Z*: edges $W-Z$ and $H^{(1)}-Z, \dots, H^{(h)}-Z$. It is a tree, so exact inference is available in closed form:

$$\hat{p}(W_t = w \mid w_{<t}) \propto e^{b_w} \sum_a e^{\mathbf{S}_{w,a}} \mu_t(a), \quad \mu_t(a) = \prod_c \sum_{j \in \mathcal{D}_t} e^{B_{j,a}^{(c)}}$$

This is fully graph-faithful (more so than MFVI), costs the same $O(h t d + |\mathcal{V}|d)$, and has an instructive form: a *mixture of exponentials over labels*, i.e. a mixture-of-softmaxes head. The exact readout therefore escapes the rank- d softmax bottleneck that the mean-field readout $\exp(\mathbb{E}_{Q_Z}[\mathbf{S}_{w,\cdot}])$ suffers (Jensen: MFVI commits to a soft single label; exact inference mixes over labels). It also replaces two Jensen-gapped messages at once: the head message becomes a per-label log-sum-exp instead of one shared attention, and the word message becomes $\log \sum_w e^{b_w + \mathbf{S}_{w,a}}$ instead of the mean $\tilde{s}_a$.

Recommendation: implement the MFVI readout as the mainline — it is the one matching Check 3 to the letter and preserving the attention correspondence — and ablate the exact readout. Both live entirely inside the graph. (*This recommendation is inverted in Part IV, Section 23.3, where the exact readout is shown to break none of the properties that motivated caution.*)

18 The six checks

Check 1 — graph-faithfulness

Every operation, with the factor it implements:

Operation	Factor / message
attention logits $F_c(j) = \sum_a Q_Z(a) B_{j,a}^{(c)}$	message $Z \rightarrow H^{(c)}$ through the arc factor, contracted with $\bar{q}_j$
attention output $\sum_j Q_c(j) B_{j,a}^{(c)}$	message $H^{(c)} \rightarrow Z$ through the same factors
query initialisation $\tilde{s}$	message $W \rightarrow Z$ through the word-label factor under $Q_W^{(0)}$
LM logits $b_w + \sum_a Q_Z(a) \mathbf{S}_{w,a}$	message $Z \rightarrow W$ through the word-label factor, plus the word unary
output softmax	the MFVI update of the variable W_t

The parameter list is exactly the factor list: $\mathbf{S}$, $\{\mathbf{T}^{(c)}\}$ (or $\mathbf{U}$, $\mathbf{V}$), r , b . There is no matrix in the computation that is not a factor, and no operation that is not a message.

The three specific objections are answered structurally: the readout now reads the label of the *unknown* slot, which has no unary to contaminate it; the path $\bar{q}_{\leq t-1} \xrightarrow{\text{arcs}} Z_t \xrightarrow{\mathbf{S}} W_t$ exists in the graph; and the “projection” is the very same factor that serves as the unary once the word is observed.

Check 2 — causality

Q_W^{readout} at slot t is a function of the parameters $(b, \mathbf{S}, r, \mathbf{T})$ and of $\{B_{j,\cdot}^{(c)}\}_{j < t}$, which are functions of $\bar{q}_{1:t-1}$, which by the induction of Part II are functions of $w_{1:t-1}$ only. The predictive label Z_t has no unary, so w_t enters nowhere. At training time the query slot t attends with the mask $\{\text{ROOT}, 1, \dots, t-1\}$ over content states that are themselves causal — and note this is the *same* strict lower-triangular mask the content stream already uses (content position i attends to $j < i$; the self-contribution enters through the unary). No leakage path exists.

Check 3 — per-step energy

$E_t^{\rightarrow}$ is written above, and the three updates are exactly its centred gradients under fixed positive weights — nothing deleted by hand. The acausal term still cannot arise, for the identical structural reason as in Part II: factors in which slot- t quantities serve later slots live inside $p(W_u, Z_u, H_u \mid z_{<u})$ for $u > t$, where $\bar{q}_t$ is a conditioning constant, not a variable of E_u . Within the slot, the $W \leftrightarrow Z$ coupling is two-sided — both are live present-time variables — so Schwarz is satisfied, not violated. Theorem 5.2 is respected exactly as before: no single global energy exists, and cannot; the global object is the directed generative chain.

Check 4 — attention correspondence and cost

With $\mathbf{T}^{(c)} = \mathbf{U}^{(c)}\mathbf{V}^{(c)\top}$, the logits are $F_c(j) = (Q_Z \mathbf{U}^{(c)}) (\bar{q}_j \mathbf{V}^{(c)})^\top$ for $j \geq 1$, plus the root logit $\langle Q_Z, r^{(c)} \rangle$ (the attention sink). This is masked scaled dot-product attention with queries from the query-stream state and keys = values = $\bar{q}_j \mathbf{V}^{(c)}$ *from the same cache the content stream uses*: the KV cache survives and is *shared* between the two streams, at zero extra memory.

What this amounts to is precisely two-stream attention (the content stream and query stream of XLNet) — here *forced by the graph* rather than posited: the content stream is the posterior label of a known word (with unary), the query stream is the prior-predictive label of the unknown slot (no unary).

Cost. Per layer: two masked-attention passes with shared K/V, i.e. a constant factor ≈ 2 on attention and $O(1)$ extra per position. The single-pass layer-parallel schedule survives: query slot t and content position t both need only content $j < t$, so all $2n$ inference problems per layer run in one pass under one triangular mask with two query tensors. The cleanest semantics is to run the content stream for T layers and then τ query-stream rounds against the final $\bar{q}$ ’s; the XLNet-style per-layer interleaving is the layer-parallel relaxation of that, exactly analogous to the relaxation already accepted for the content chain. RPE: distances $t - j \geq 1$, same clipped table, no change.

Check 5 — training

The loss is the model’s own negative log-likelihood, and nothing else:

$$\mathcal{L} = - \sum_{t=1}^n \log Q_{W_t}^{\text{readout}}(w_t)$$

with $\lambda_W = 1$, so that the readout is an untempered conditional. (Any other λ_W is a temperature; with b and $\mathbf{S}$ learned it is absorbable, so fix it to 1.) EOS is a vocabulary item. Slot 1 predicts w_1 from ROOT alone ($\mathcal{D}_1 = \{\text{ROOT}\}$), so the model has a proper first-word distribution without a BOS hack.

Gradients. “Frozen” means constant with respect to slot- t ’s inner inference, implemented as ordinary forward values — so $\partial\mathcal{L}/\partial\theta$ flows through the readout into Q_Z^{pred} , through the attention into $B_{j,a}^{(c)}$, into $\bar{q}_j$, and back through the entire prefix computation, exactly as with cached activations in a transformer. The matrix $\mathbf{S}$ receives gradient in both of its roles — unary of observed words, and emission of predicted words — which is tied-embedding training, forced by remark (b) of Section 12.

Teacher forcing. All n query slots and all n content positions are computed in one parallel pass as in Check 4. Nothing about the established single-pass training changes except the second query tensor.

Check 6 — generation, and the consistency question

The loop.

1. Hold $\bar{q}_{1:t-1}$ and the shared KV cache.
2. Run the predictive slot: τ inner rounds, then the W -readout.
3. Sample $w_t \sim Q_W$.
4. Run the *observed* step for slot t : the same energy with Q_W clamped to δ_{w_t} , i.e. the step of Part II, producing $\bar{q}_t$; freeze it and append $\bar{q}_t \mathbf{V}^{(c)}$ to the cache.
5. $t \rightarrow t + 1$.

Per generated token this is two inner problems over the cache instead of a GPT’s one: a constant factor, no asymptotic change.

The consistency issue, stated plainly. The predictive belief $Q_Z^{\text{pred}} \approx p(Z_t \mid w_{<t})$ and the frozen $\bar{q}_t \approx p(Z_t \mid w_{\leq t})$ are *the prior-predictive and the posterior of the same variable* — two inference queries on one slot, not two models. Their difference is Bayes conditioning, visible exactly in the exact-inference forms of Section 17.2:

$$Q_Z^{\text{pred,exact}}(a) \propto \left[\sum_w e^{b_w + \mathbf{S}_{w,a}} \right] \mu_t(a), \quad \text{versus} \quad \bar{q}_t^{\text{exact}}(a) \propto e^{\mathbf{S}_{w_t,a}} \mu_t(a),$$

the same context factor μ_t with the word-factor message swapped from marginalisation to evidence. So this is not a bug, and the chain must freeze the *posterior* version — it is what carries the observed word into the future; the predictive beliefs are transient query answers, legitimately superseded.

Two genuine residues remain, and they should be in the write-up rather than discovered by a reviewer.

Caveat 18.1 (Approximation-level divergence). Under MFVI the two queries do not share μ_t : the attention is re-run with a different query, so the pattern used to *predict* w_t and the pattern used to *encode* w_t can differ more than exact inference would allow. Legal mitigations: warm-start the observed step’s $\{Q_c\}$ from the predictive step’s (initialisation is a free choice of the inner scheme), or share μ_t outright by using the exact tree readout of Section 17.2.

Caveat 18.2 (Modelling-level omission, already paid for). Exact filtering would let the observed w_t revise the *prefix* labels: w_t is evidence about $z_{<t}$ through the new word factor and Z_t ’s arcs (explaining away). Frozen-prefix assumed-density filtering drops exactly this backward evidence. It is the same term the chain of Part II already drops (there the unary was the evidence), it is the price of the KV cache, and nothing new is lost — but it should be stated.

19 What the construction buys

The output mechanism is now made entirely of pre-existing parts:

- the LM softmax is the MFVI update of a graph variable;
- the output matrix is the unary factor read in the other direction — weight tying as a theorem;
- the output bias is a word unary;
- the [MASK]-style learned query is the prior word message;
- two-stream attention is the prior/posterior distinction on one slot;
- the mixture-of-softmaxes readout is exact inference on the slot tree.

Each of these is a known engineering practice that the model *derives*. That, and not any single equation, is the answer to “white-box decoder or probabilistic model with a neural head stapled on”: there is no longer any operation between corpus and loss that is not a message through a declared factor.

20 The honest ledger

Each item below is taken up and resolved, repaired, or reframed in Part IV.

1. **Jensen gap and rank.** The mean-field readout carries a Jensen gap against the model’s true predictive distribution and is rank-limited by d . The exact tree readout (Section 17.2) fixes both at equal cost, but steps outside the pure-MFVI computation pattern. Run it as an ablation and report which is used.
2. **Constant-factor cost.** Training and generation cost roughly $2\times$ the attention of a GPT of equal depth, with a shared cache — the price of separating the two roles that a GPT’s hidden state conflates.
3. **The label bottleneck.** All prediction flows through the d -dimensional discrete label bottleneck by construction. On language-modelling benchmarks this, rather than the mechanism above, is where capacity pressure will show first. Say so pre-emptively, with the exact readout and d -scaling as the available levers.

A diagnostic worth running first. On a small model, verify that Q_Z^{pred} and $\bar{q}_t$ correlate the way a prior and a posterior should: close on predictable tokens, divergent on surprising ones. If they do not, the inner-loop approximation — not the design — is where to look, and that diagnostic is worth having in hand before presenting the construction.

Part IV — The defect analysis

21 Overview: two earlier recommendations get inverted

Parts I–III closed with an honest ledger. Working its items to the bottom, rather than acknowledging them again, changes two of the document’s own earlier recommendations, and it is better to say so at the top than to let a reader discover it: (i) the exact tree readout

of Section 17.2 should be the *mainline*, not the ablation — it breaks none of the properties that motivated caution, and it removes the query stream, taking most of Defect 3 with it (Section 23.3); (ii) the label bottleneck, once quantified, is not located where the ledger placed it — the readout bottleneck is identical in kind and degree to the baseline’s own softmax bottleneck at matched width, and the genuine risk sits one level upstream, in the capacity of the inference dynamics (Section 22). Each defect below ends in either a concrete fix with its checks carried out, or a negative result with the framing to publish.

22 Defect 1: the label bottleneck

22.1 Quantification: three bottlenecks, disentangled

The phrase “label bottleneck” conflates three different constraints. They have very different severities.

The readout bottleneck.

Proposition 22.1 (Readout expressivity). *Fix any set of contexts and let A be the matrix whose rows are the log-conditionals $\log p(\cdot \mid \text{context})$ over the vocabulary.*

- (i) *Under the mean-field readout, $\ell = b + \mathbf{S}Q_Z$ with $Q_Z \in \Delta^{d-1}$, the rows of A lie in an affine subspace of dimension at most d .*
- (ii) *Under a linear-softmax head of width m (logits Eh , $h \in \mathbb{R}^m$, plus bias), the rows lie in an affine subspace of dimension at most $m + 1$.*
- (iii) *The exact tree readout $\hat{p}(w) \propto e^{bw} \sum_a e^{\mathbf{S}^{w,a}} \mu_t(a)$ is not an affine family: if the component distributions $p(\cdot \mid a) \propto e^{b+\mathbf{S} \cdot a}$ are not all identical, the log-mixture map $\mu \mapsto \log \hat{p}(\cdot)$ is nonlinear in its parameters, and its image is a curved $(d-1)$ -dimensional manifold that is not contained in the subspace of (i).*

Proof. (i) $\{\mathbf{S}q : q \in \Delta\}$ lies in the affine hull of the columns of $\mathbf{S}$, of dimension at most $d - 1$; adding the fixed b preserves the affine set; the per-row normaliser adds one dimension along $\mathbf{1}$. (ii) identically, with a linear subspace of dimension m in place of the hull. (iii) log of a non-trivial mixture is not affine in the mixture weights (e.g. $\log(\frac{1}{2}p_1 + \frac{1}{2}p_2) \neq \frac{1}{2}\log p_1 + \frac{1}{2}\log p_2$ pointwise unless $p_1 = p_2$); a family affine in no parametrisation of an affine subspace cannot lie in one of dimension d while containing the d component vertices and their non-affine combinations. We claim only escape from the constraint of (i), not full-dimensionality of the affine hull. $\square$

Consequence, stated bluntly: **the mean-field readout’s bottleneck is the classical softmax bottleneck (Yang et al. 2018), in identical kind and degree to a baseline transformer’s own output head at $d = d_{\text{model}}$.** The comparison unit is $d \leftrightarrow d_{\text{model}}$, and the PT paper already runs at parity ($d = 384$ against transformer widths 256–512). Parity of the readout requires $d \approx d_h$, full stop. The exact readout is a different family altogether: a mixture over d *fixed* component softmaxes with context-dependent weights — weaker than a full mixture-of-softmaxes head (whose components are context-dependent), but strictly outside the linear family at zero parameter cost.

The chain bottleneck. $W_t \perp w_{<t} \mid z_{<t}$ compresses the prefix into $(\bar{q}_1, \dots, \bar{q}_{t-1})$ — one point of Δ^{d-1} per position, consumed *linearly* by the future through $B_{j,a}^{(c)} = \sum_b \bar{q}_j(b) \mathbf{T}_{a,b}^{(c)}$. A transformer’s KV cache is one point of $\mathbb{R}^{d_h}$ per position, consumed linearly through the K/V projections. Same structure; the per-position width is $d - 1$ against d_h . The two genuine

differences: (a) the simplex geometry — bounded, non-negative, summing to one: a bounded-activation network; (b) one scale coupling created by weight tying: the output logit range equals the scale of $\mathbf{S}$, which is shared with the unary. The in-model lever for (b) is λ_W : the entropic Frank–Wolfe weight of the W variable is a legitimate per-variable message weight (the paper already treats λ_H as a hyperparameter), and $\lambda_W < 1$ decouples output sharpness from unary scale. This supersedes the flat “fix $\lambda_W = 1$ ” of Part III; the trade is stated rather than hidden: with $\lambda_W \neq 1$ the readout is a tempered conditional, and the “calibrated by construction” claim weakens to “calibrated up to one declared temperature”.

The dynamics bottleneck — the real one. The information-theoretic statement requested: at slot t the predictive distribution factors through d non-negative numbers ($\mu_t(\cdot)$, or Q_Z), exactly as a GPT’s factors through d_h numbers. *The bottleneck dimensionality is matched, so the risk is not the pipe but the pump*: whether PT’s inference dynamics — attention-only, no feed-forward structure, parameters shared across iterations, simplex-constrained states — can deposit as much usable information into those d numbers as a GPT’s stack deposits into d_h . The paper itself already identified this at scale (§6: underperformance on large data, suspected cause the missing feed-forward structure). This is where the experiment will fail if it fails, and it is addressable in-graph (Section 22.2).

22.2 Widening the model without leaving the graph

Raise d . Graph-faithful trivially. The cost is what raising d_{model} costs: $\mathbf{S}$ is $|\mathcal{V}| \times d$ (the embedding cost every LM pays), decomposed $\mathbf{T}$ is linear in d , attention flops are governed by $r \cdot h$, the readout is $O(|\mathcal{V}|d)$. This is the default lever; its real price is the one Defect 1(c) names: a d in the thousands erodes “small interpretable label set”.

Factored labels — the flagship option. Take K label variables $Z_t^{(1)}, \dots, Z_t^{(K)}$ per position, each over d' values, with K *binary* word–label factors sharing W_t ,

$$\phi(W_t, Z_t^{(k)}) = \exp(\mathbf{S}_{W_t, Z_t^{(k)}}^{(k)}), \quad k = 1, \dots, K,$$

and each head channel c assigned a child component $k(c)$ and a head component $k'(c)$, so that $B_{j,a}^{(c)} = \sum_b \bar{q}_j^{(k'(c))}(b) \mathbf{T}_{a,b}^{(c)}$. The predictive per-step energy is the obvious sum,

$$E_t^{\rightarrow} = - \sum_w Q_W(w) b_w - \sum_k \sum_{w,a} Q_W(w) Q_{Z^{(k)}}(a) \mathbf{S}_{w,a}^{(k)} - \sum_c \sum_{j \in \mathcal{D}_t} \sum_a Q_c(j) Q_{Z^{(k(c))}}(a) B_{j,a}^{(c)},$$

multilinear, with honest gradients per variable. Checks: causality untouched (nothing new sees the future); attention correspondence per channel with component-specific queries and keys; cost linear; parameters $|\mathcal{V}|Kd' = |\mathcal{V}|D$ at total width $D = Kd'$ — identical to a flat label of width D . What factoring actually buys: the slot graph is still a tree (W_t joined to K stars), so the exact readout becomes a **product of K mixtures**,

$$\hat{p}(w) \propto e^{b_w} \prod_{k=1}^K \sum_{a=1}^{d'} e^{\mathbf{S}_{w,a}^{(k)}} \mu_t^{(k)}(a), \quad \mu_t^{(k)}(a) = \prod_{c: k(c)=k} \sum_{j \in \mathcal{D}_t} e^{B_{j,a}^{(c)}},$$

a strictly richer family than one mixture at the same D , and the strongest graph-faithful head available here. Interpretability *improves*: K simultaneous discrete annotations per word (a feature bundle rather than a monolithic tag), and channels typed by which component they read. Stated honestly: under the *mean-field* readout, factoring buys nothing over flat width D (logits $= b + \sum_k \mathbf{S}^{(k)} q^{(k)}$, affine dimension $\leq D$); the gain exists only through the exact readout. Per-channel labels are the special case $K = h$, $k(c) = c$; there is no reason to hard-couple K to h .

Continuous labels. Gaussian slot variables give a closed-form readout $\log \hat{p}(w) = s_w^\top m + \frac{1}{2} s_w^\top \Sigma s_w + \text{const}$ (the Gaussian moment generating function): a log-quadratic head with per-word variance sensitivity, and an unbounded state that dissolves the scale tension. But it trades away the discrete interpretability that is the thesis and rewrites the inference mathematics (linear-Gaussian message passing). A different project; one paragraph of future work, not this design.

The graph-faithful feed-forward. The upstream capacity fix already exists inside the paper: the global variables of Appendix B.3. Add to each slot’s conditional a global-head variable G_t (single-split form) with the binary factor $\exp(B'_{m,Z_t})$. The slot stays a tree (G_t is one more leaf off Z_t ; μ_t gains the multiplicative term $\sum_m e^{B'_{m,a}}$); causality holds (position-local); the mean-field update gains exactly the GFU term $\sigma(qB'^\top)B'$ that the paper shows is the feed-forward analogue; cost $O(md)$ per position. **This is the principled answer to “PT lacks an FFN”, and it should ship in the causal model from day one** — otherwise the experiment confounds “causal construction” with the known encoder-side capacity gap.

The tripwires. What quietly reintroduces a neural head: any MLP from Q_Z to logits; any context-dependent $\mathbf{S}$ (a hypernetwork is not a factor); any nonlinearity applied to $\bar{q}$ between steps outside the update equations; per-iteration untied parameters — each iteration then performs inference in a *different* model, which is amortised inference in probabilistic clothing; cross that line only explicitly and with its status declared. The clean boundary statement:

In this model class, expressivity is bought by adding *variables and factors*,
never by adding *maps*.

Every legitimate widening above is a variable ($Z^{(k)}$, G_t) or a weight (d , λ_W); everything illegitimate is a function.

22.3 The honest framing

At matched width, with the exact readout and the B.3 globals, there is no *a priori* reason the construction must lose: the readout family is not weaker than the baseline’s, the bottleneck width is matched, and the known capacity gap has an in-model repair. If a gap persists, it is attributable to the simplex geometry, the iteration-tied parameters, and discreteness — and then the honest, publishable framing is exactly the one the ledger anticipated: a white-box decoder with a *measured* interpretability/perplexity trade curve over (d, K) , where the interpretability axis is itself quantified (probing $\bar{q}$ for POS/dependency content, in the spirit of the paper’s Appendix F). One sentence to pre-empt the reviewer: the model’s per-token state is a probability vector by construction, so every capacity comparison is at bounded activations; report the trade curve, not a single point.

23 Defect 2: the prior/posterior divergence under MFVI

23.1 A contraction bound

Within slot t , the predictive and observed runs differ only in the word message m_W : $\tilde{s}$ against $\mathbf{S}_{w_t,\dots}$. Both share the constants $B^{(c)}$ and the schedule.

Lemma 23.1 (Divergence bound). *Run the asynchronous inner loop (H -update from the current q , then Z -update) in both modes, and let δq_s , $\delta \alpha_s^{(c)}$ denote the norms of the differences*

after round s (Euclidean norms on centred vectors; shifts are irrelevant by softmax invariance). Then

$$\delta\alpha_s^{(c)} \leq \frac{\|B^{(c)}\|_2}{2\lambda_H} \delta q_{s-1}, \quad \delta q_s \leq \frac{\|\Pi \Delta m_W\|}{2\lambda_Z} + \rho \delta q_{s-1}, \quad \rho := \frac{\sum_c \|B^{(c)}\|_2^2}{4\lambda_Z\lambda_H},$$

and if $\rho < 1$,

$$\delta q_\infty \leq \frac{1}{1-\rho} \cdot \frac{\|\Pi \Delta m_W\|}{2\lambda_Z}, \quad \Pi \Delta m_W = \Pi(\mathbf{S}_{w_t, \cdot} - \mathbb{E}_{Q_W} \mathbf{S}_{w, \cdot}).$$

Proof. The softmax Jacobian satisfies $\|\text{diag}(p) - pp^\top\|_2 \leq \frac{1}{2}$: for unit v , $v^\top(\text{diag}(p) - pp^\top)v = \text{Var}_p(v) \leq \frac{1}{4}(\max_i v_i - \min_i v_i)^2 \leq \frac{1}{2}\|v\|^2$ (Popoviciu, then $(a-b)^2 \leq 2(a^2 + b^2)$). The H-update reads q_{s-1} : by the mean value theorem, $\delta\alpha_s^{(c)} \leq \frac{1}{2\lambda_H} \|B^{(c)}\| \delta q_{s-1}$. The Z-update reads $m_W + \sum_c B^{(c)\top} \alpha_s^{(c)}$, so its input difference is at most $\|\Pi \Delta m_W\| + \sum_c \|B^{(c)}\| \delta\alpha_s^{(c)}$, and applying the softmax Lipschitz constant $\frac{1}{2\lambda_Z}$ and substituting gives the recursion. Unroll the geometric series. $\square$

Read the kick: $\Pi \Delta m_W$ is the **embedding-space surprisal of the observed word** — the divergence between the prediction-time and encoding-time attention patterns is provably small on predictable tokens and grows with surprise, which is exactly the prior/posterior behaviour Bayes prescribes. Honest limits of the bound: $\|B^{(c)}\|_2 \leq \|\mathbf{T}^{(c)}\|_2 \|\bar{Q}\|_2$ can grow like $\sqrt{t}$ in the worst case (the standard operator-norm price over growing attention), so ρ should be read as governed by $\|\mathbf{T}\|^2/(\lambda_Z\lambda_H)$ — which the paper’s own machinery already regulates (the variance-calibrated λ ’s of Appendix A.5; the L2 penalty on $\mathbf{T}$ used for MLM). No clean bound in the entropy of Q_Z^{pred} exists without further assumptions, and none is manufactured here. The qualitative corollary matters most: $\rho \geq 1$ is precisely the regime in which the inner loop admits multistability — prediction and encoding landing in *different fixed points* — so the contraction condition is not decoration; it is the boundary of the feared pathology.

23.2 Sharing the attention pattern is legal — it is a warm start

The decisive observation: $\partial E_t / \partial Q_c(j) = -\sum_a Q_Z(a) B_{j,a}^{(c)}$ contains no word message — **the H-update functional is identical under $E_t^\rightarrow$ and E_t** ; the two runs’ attentions differ only through the q at which they are evaluated. Therefore “compute the attention once per slot and reuse it for both queries” is exactly: *initialise the observed slot at the predictive slot’s state* (initialisation is free in MFVI), *then run one H-round and one Z-round of E_t* . Every update in that procedure is an honest gradient of E_t ; Check 3 is satisfied to the letter; nothing is clamped or forced. The trade, stated: $\bar{q}_t$ is then a one-round approximation of the observed fixed point, biased toward the prior; the inner count τ_{obs} interpolates continuously between shared- μ ($\tau_{\text{obs}} = 1$) and fully re-solved.

23.3 Inverting the recommendation on the exact readout

What the exact tree readout of Section 17.2 actually breaks was checked item by item. The answer: none of the three things that would have justified keeping it as the ablation.

Layer-parallel schedule: unaffected — the readout is closed-form on top of the final $\bar{q}$ ’s. In training, $\log \mu_t(a) = \sum_c \text{LSE}_{j < t} B_{j,a}^{(c)}$ (seeded with $r_a^{(c)}$) is a causal prefix log-sum-exp along the sequence: one **logcumsumexp** scan, $O(ndh)$, fully parallel.

KV cache: the content stream’s cache is untouched, and the readout acquires its own μ -**cache**: a running $d \times h$ table updated by one **logaddexp** per generated token, $O(dh)$ — cheaper than any attention pass.

Attention correspondence: the content stream’s is untouched. The readout’s changes shape from softmax attention to per-label LSE pooling, whose gradient is a softmax over positions *per label per channel* — i.e. one obtains $d \cdot h$ inspectable attention maps instead of h , plus the per-label context evidence $\log \mu_t(a)$ and per-word label responsibilities. The interpretability of the prediction step goes up, not down.

And it **removes the query stream entirely** — no predictive inner loop, no Q_Z^{pred} iterations — which is most of Defect 3.

The honest costs, so that nobody breaks this in one question later: (1) Check 3 must be restated, from “every update is a softmax of a gradient” to “*every step is valid inference in the declared model* — MFVI where exactness is impossible (the chain), exact sum–product where the subgraph is a tree (the slot)”. That is a strengthening in substance and a change in letter; the paper’s own B.3.1 (exact collapse of fixed-distribution neighbours) is the precedent. (2) The optimisation behaviour of LSE-pooling gradients at LM scale is untested, whereas softmax attention’s is battle-hardened; and the $|\mathcal{V}| \times d$ readout is an LSE rather than a matmul — the same flops with worse hardware constants (element-wise exponentials); chunk it over the vocabulary like a fused cross-entropy. (3) The single summary belief Q_Z^{pred} is lost; μ_t replaces it (finer, but different).

Verdict, inverted as the question anticipated: exact readout mainline; mean-field two-stream as the ablation. A pleasant corollary: with the exact readout, Defect 2 survives only in the mild warm-start form of Section 23.2.

24 Defect 3: the $2\times$ cost

24.1 Is the second stream avoidable?

Two answers. Under the new mainline: **yes, trivially** — **the query stream no longer exists**; the per-token cost is one content-stream solve, an $O(dh)$ μ -cache update, and the vocabulary readout. Under the mean-field design: no exact algebraic recovery exists — q^{pred} and q^{obs} are fixed points of the slot map at two different word messages, the map is a composition of softmaxes, and “subtract the unary contribution inside a softmax” is undefined; the nonlinearity is indeed fatal to anything exact. What does exist is the first-order relation $q^{\text{obs}} \approx q^{\text{pred}} + J \Delta m_W$, whose honest implementation is precisely the warm start of Section 23.2: the observed solve collapses to one (H, Z) round from the predictive state. Even in the mean-field design the realistic overhead is $\sim (1+\varepsilon)$ streams, not 2.

24.2 Realistic accounting

Per token and per layer, with the usual calibration $rh \approx d$ for the decomposition:

Architecture	FLOPs per token per layer (approx.)
Transformer layer	$4d_h^2$ (QKVO) + $8d_h^2$ (FFN) + $2nd_h$ (scores, mix) $\approx 12d_h^2 + 2nd_h$
PT content layer	$3drh$ ($\bar{q}\mathbf{V}$ once — keys are values — plus $q\mathbf{U}$ and the $\mathbf{U}^\top$ output) + $2nrh \approx 3d^2 + 2nd$
PT, mean-field two-stream	$\approx 5d^2 + 4nd$ (keys shared)
PT, exact-readout mainline	content layer only; plus per token $O(dh)$ for the μ -scan and $O(\mathcal{V} d)$ for the readout

Even the *two-stream* mean-field version sits at roughly $0.4\text{--}0.6\times$ of a transformer layer, because the FFN — which dominates a transformer layer — is absent. If the B.3 globals are added (as Section 22.2 argues they should be), add $\approx 2md$ per token; the gap narrows but does not close. Two honest footnotes: PT shares parameters across iterations, so parameter-matched

comparisons let PT run more iterations per parameter — decide and declare whether depth or parameters are matched; and the paper’s measured $\sim 3\times$ wall-clock slowdown comes from the RPE table gathers (its Appendix C), an engineering axis orthogonal to stream count. “ $2\times$ attention” was the wrong headline; the right one is: *attention-only layers, no FFN, roughly half a transformer layer’s flops — plus a vocabulary readout every position, which every language model pays*.

25 Defect 4: explaining away

25.1 It is the same term: a proof by directedness

Proposition 25.1 (Equivalence of the dropped evidence). *The evidence discarded by frozen-prefix filtering in the generative chain of Part III is, term for term, the evidence discarded in the conditional chain of Part II. Promoting the words to variables creates no additional dropped pathway.*

Proof. Slots are ordered so that slot u ’s variables are descendants of $z_{<u}$. Two cases. *Unobserved future* (prediction time): marginalising an unobserved descendant block integrates its conditional to 1, so future slots are inert and $p(z_{<t} \mid w_{<t})$ is untouched by their existence — nothing is dropped that exact inference would keep. *Observed future word w_u* : the evidence about $z_{<u}$ flows through exactly one gate — the word-label factor with W_u clamped, i.e. the unary $\exp(\mathbf{S}_{w_u, Z_u})$ (Part III, remark (a)), then out through Z_u ’s arcs. That clamped factor *is* the unary whose backward evidence Part II drops, and no other factor touches W_u . Hence the discarded terms coincide. $\square$

A consistency check across the document: what is dropped is exactly the “legal within-prefix term 2” — the thing Proposal C retains. The three parts tell one story.

25.2 Measuring the cost

Proposal C is the gold standard by construction: on short sequences, compare frozen-chain beliefs against per-prefix re-inference — report $\text{KL}(\bar{q}_j^C \parallel \bar{q}_j^{\text{frozen}})$ as a function of position and of distance from the sequence end, and, decisively, the next-token NLL gap between predicting from frozen and from re-inferred prefixes: that gap *is* the price of the KV cache in nats per token. Run it specifically on right-context-disambiguation material (garden-path sentences, late-disambiguating relative clauses), where the dropped revision is linguistically load-bearing; a uniform corpus average will understate it.

25.3 A cheap partial repair that stays inside the graph: fixed-lag smoothing

When w_{j+1} is observed, the exact one-step backward message to z_j (with the rest of the prefix held at its beliefs, and the slot marginalised exactly — the slot is a tree) is the likelihood

$$\ell_j(b) = \frac{\sum_a e^{b_{w_{j+1}} + \mathbf{S}_{w_{j+1}, a}} \mu_{j+1}(a; b)}{\sum_{w'} \sum_a e^{b_{w'} + \mathbf{S}_{w', a}} \mu_{j+1}(a; b)}, \quad \mu_{j+1}(a; b) = \prod_c \left[\sum_{k \in \mathcal{D}_{j+1}, k \neq j} e^{B_{k,a}^{(c)}} + e^{\mathbf{T}_{a,b}^{(c)}} \right],$$

and the refresh is $\bar{q}_j(b) \leftarrow \bar{q}_j(b) \ell_j(b)$, renormalised — a single exact backward sum-product step in the directed model. Extending to lag k refreshes the last k positions per step. Checks: it is inference over existing factors only (the same restated Check 3 status as the exact readout); causality *in the language-model sense* is preserved — $\bar{q}_j$ now depends on $w_{1:j+k}$, all of which precede every prediction that will consume it; nothing downstream of any prediction target

leaks. Costs, stated: $O(d^2h)$ per refreshed position — the b -dependence of μ does not factor through the low-rank $\mathbf{T}$, and no pretence is made that it does; truncating $\bar{q}_j$ to its top- m support cuts this to $O(mdh)$ — and the KV cache becomes lag- k mutable: the last k rows are rewritten each step, $O(kdrh)$ per token. This is assumed-density filtering upgraded to a fixed-lag smoother — textbook-principled — with the knob k running from 0 (the current design) to t (Proposal C). Whether to pay for it is exactly what the diagnostic above decides.

26 The first experiment: a localisation ladder

The prior/posterior correlation check proposed at the end of Part III is a *unit test*: it verifies inference health and diagnoses Defect 2, which the redesign above has demoted. It is not the experiment that decides viability. The one that does — the one whose outcome most changes the design — is a localisation ladder on a small corpus (PTB or BLLIP-XS, where the paper already has encoder numbers), all rungs sharing one codebase and matched width $d = d_h$:

1. a full GPT baseline;
2. a GPT handicapped to PT’s structural budget: no FFN, shared layers, tied embeddings, $K=V$ tied per the Section-3 correspondence;
3. the causal PT with the exact readout (the mainline of Section 23.3);
4. rung 3 minus each ingredient in turn: mean-field readout instead of exact; without the B.3 globals; factored labels $K \in \{1, 2, 4\}$ at fixed total width.

The two decisive gaps. **(2) – (3)** is the price of *this construction* — of simplex states, discreteness, and inference-as-computation — with the known FFN/sharing handicap already removed from the comparison. If it is small, the programme is safe, and the residual gap to (1) is the encoder paper’s known capacity story with an in-graph fix. If it is large, the label dynamics are the binding constraint, and the (d, K) scaling study becomes the thesis. And **(exact) – (MF)** within rung 4 measures, in nats, how much the mixture family buys — simultaneously the sharpest empirical statement about Defect 1’s readout component. The correlation check and the frozen-versus-re-inferred probe of Section 25 ride along as cheap instruments *inside* this experiment; neither should run first on its own. One practical warning for rung 3: chunk the LSE readout over the vocabulary like a fused cross-entropy, or the memory of the $|\mathcal{V}| \times d$ intermediate will dominate everything else being measured.

27 Summary

1. **Part I.** Causality, a single global energy, and contextuality are mutually incompatible (Theorem 5.2); any two are compatible (Section 6). The obstruction is Schwarz’s theorem: every coupling in a single smooth energy is two-sided, so term 1 and term 2 are two faces of one Hessian entry. Approach A is therefore not the gradient field of any energy unless $\mathbf{T}$ is degenerate, in which case the model is non-contextual (Corollary 7.1). The theorem requires the architectural formalisation of causality: with the literal trajectory formalisation it is false (Proposition 8.1).
2. **Part II.** The minimal change of model class is a directed chain of conditional CRFs with variational filtering. The acausal message is never born; each step is an honest gradient field; the reduction to masked scaled dot-product attention is exact; the symmetry assumption on $\mathbf{T}$ becomes unnecessary; the KV cache falls out of the model. Per-prefix re-inference is the alternative subproblem choice, with smoothing semantics and factor- n cost.

3. **Part III.** Promoting words to variables makes prediction an inference query: the next-slot label has no unary and is inferred from the prefix through arc factors alone, then mapped to the vocabulary by the word-label factor. Weight tying, the learned mask embedding, two-stream attention, and a mixture-of-softmaxes readout are all derived. The prior/posterior distinction at each slot is correct Bayesian semantics, with two stated approximation-level residues.
4. **Part IV.** Quantification relocates the risk: the mean-field readout’s bottleneck equals the baseline’s own softmax bottleneck at matched width (the comparison unit is $d \leftrightarrow d_{\text{model}}$), the exact tree readout escapes it as a mixture family, and the genuine risk is the capacity of the inference dynamics — addressed in-graph by factored labels (a product-of-mixtures readout) and by the paper’s global variables as a graph-faithful feed-forward. The prior/posterior divergence obeys a contraction bound with an explicit loop gain and an embedding-surprisal kick; sharing the attention pattern is the legal warm-start schedule; the exact readout becomes the mainline and removes the query stream; the cost accounting corrects to roughly half a transformer layer; the dropped explaining-away term is proved identical to Part II’s, with fixed-lag smoothing as the graded in-graph repair; and the first experiment is the localisation ladder, not the correlation check.